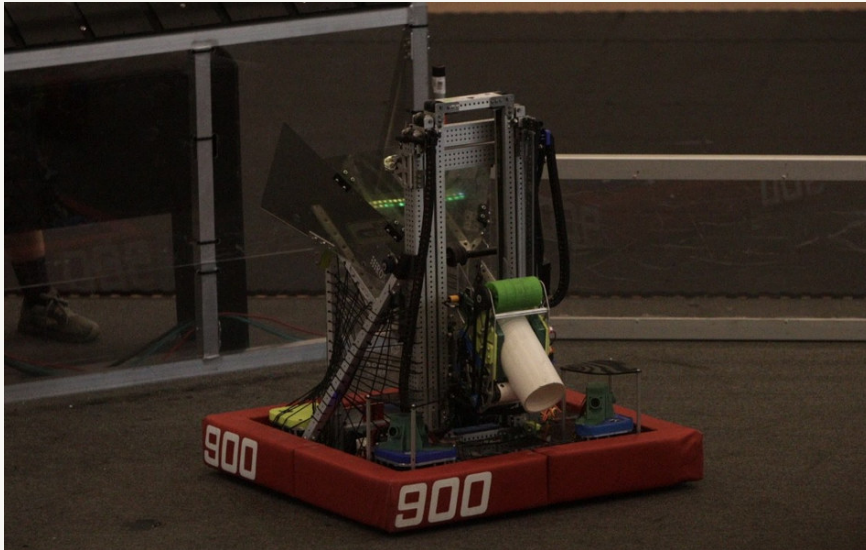


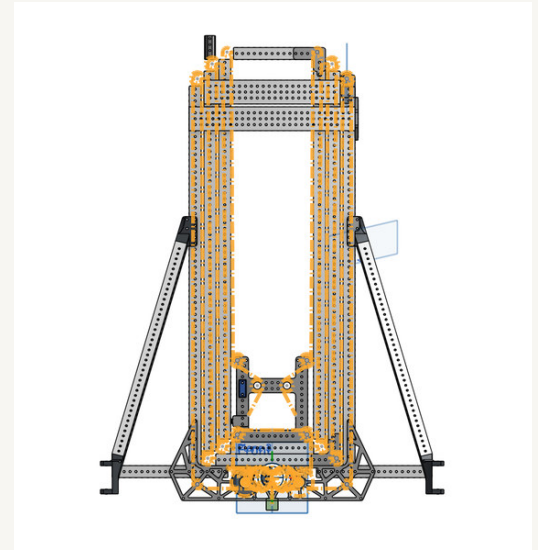
C4 competition robot

FIRST Robotics Competition | Team 900, The Zebracorns

Design and Mechanical Lead | 2025 season



BUILT ROBOT



ELEVATOR BELT ROUTING + GEARBOX

36 to 74 in

stowed / extended elevator height

<4 in

elevator thickness, down from >5.5 in

~40%

cost reduction vs. original manufacturing approach

ROLE AND DESIGN PROCESS

I led 25 mechanical students and five CAD students, designed the elevator and managed the master geometry and Onshape workspace. We left algae scoring and deep climbing out of scope to focus on tuning and driver practice. Prototyping and CAD reviews supported fabrication, integration and elevator testing.

ELEVATOR AND FABRICATION

The three-stage elevator combines machined 6061 aluminum, two Kraken X60 motors, a 3:1 gearbox and two internal HTD 5 mm belts. Master sketches defined packaging and belt paths; custom Onshape FeatureScripts generated pulleys and gears. Markforged-printed and waterjet-cut aluminum bearing blocks reduced cost by about 40% and subsystem lead time by about two weeks.

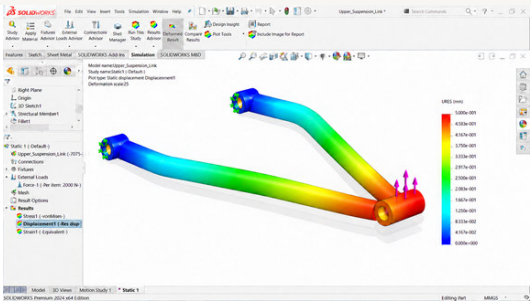
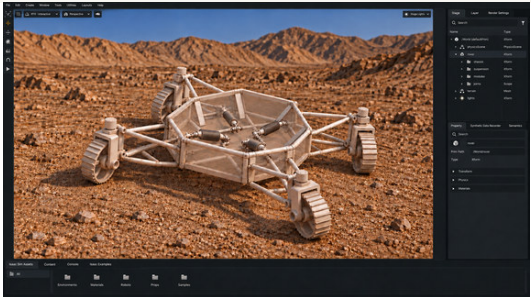
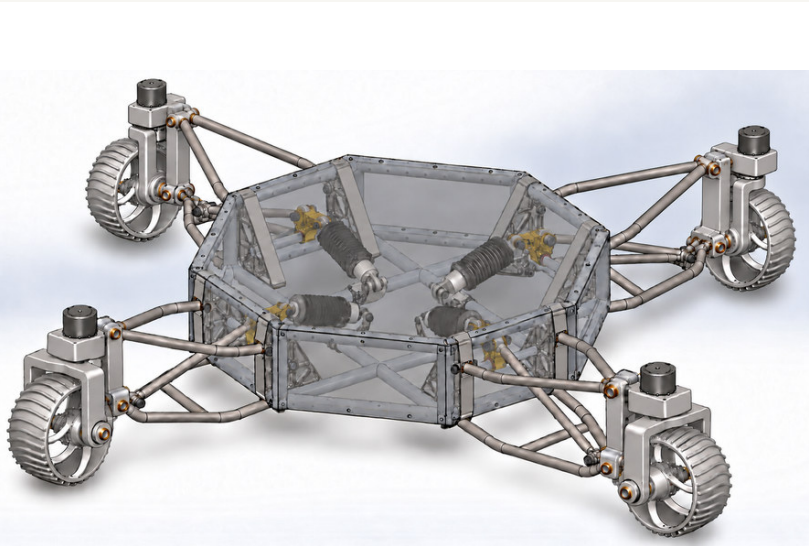
COMPETITION RESULTS

The elevator had no mechanical failures during competition. The team won the 2025 NC Championship and ranked in the top 11% of FRC teams worldwide, despite our school heavily cutting our mentor team, resources, and funding.

Suspended swerve rover

Four-wheel steering and double-wishbone pushrod suspension

Mechanical design, basic FEA and early simulation | September 2025-May 2026



+/-150 mm

wheel-travel design target

Double wishbone

pushrod suspension

+/-5 deg

camber-range design target

CAD WORK

I modeled the tubular chassis, double wishbones, pushrods and inboard suspension using master sketches, 3D sketches and weldments. Original component selections included 7075 aluminum parts, QA1 coilovers with 350 lb/in springs and AKM42C servomotors. The illustrated hub-drive layout is a later concept revision.

FUTURE SIMULATION WORK

I plan to develop autonomous navigation in Isaac Sim, including terrain mapping, waypoint following, obstacle avoidance and replanning. Edge-case scenes would cover cross-slopes, unequal wheel loading, low traction, steps near suspension travel limits and tight passages. I would compare path tracking, wheel contact/slip, steering error and chassis attitude across these conditions.

CURRENT STATUS AND NEXT STEPS

Completed basic suspension-link FEA and basic teleoperation. Next: autonomy and edge-case simulation, detailed component and assembly FEA, and topology or parametric optimization. The geometry values above remain design targets.

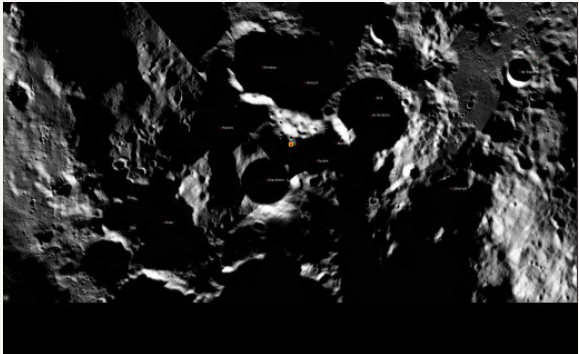
L'SPACE mission concept

NASA L'SPACE Mission Concept Academy | Team 24

Project Manager | May-August 2026 | Student design study



TEAM CONCEPT: BASE STATION + SCOUT



Station	Scout
Power, charging, thermal support and data relay	Cold-gas mobility, mapping and hazard sensing

JMARS | SHACKLETON-DE GERLACHE RIDGE

20 members

student team across science, engineering and planning

175 kg

combined launch-mass ceiling

95 days

planned surface mission length

ROLE AND REVIEW PROCESS

I managed the 20-member team, coordinated reviews and deliverables, and contributed to vehicle CAD, mechanical integration and science-to-engineering traceability. From May to August 2026, we progressed through MCR, SRR, MDR and PDR, using assigned tasks, soft deadlines, daily tracking and cross-section reviews.

MECHANICAL AND SYSTEM INTERFACES

Preliminary frame selections were 6061-T6 for the station and 7075-T6 for the scout. The concept included docking alignment, spring-probe charging, a local UHF link and X-band direct-to-Earth communications. Depth and inertial sensing supported terrain mapping. Four planned scout excursions fit within the 95-day surface mission.

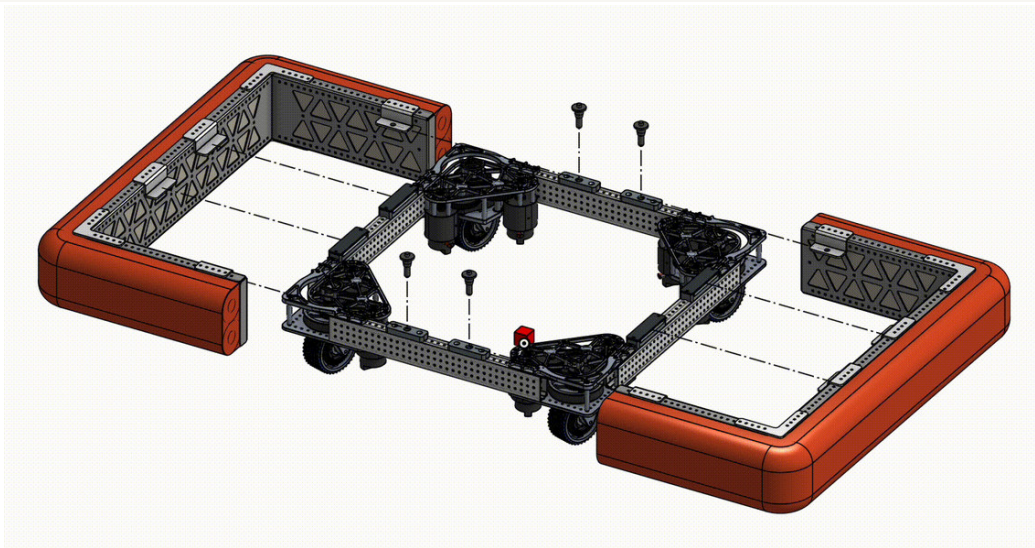
DELIVERABLE AND STATUS

Completed a student Preliminary Design Review with vehicle CAD, science-to-requirement traceability, N-squared interface diagrams, FMEA and verification plans. Propulsion sizing, integrated mass/power/thermal budgets and hardware validation remained open.

Configurable sheet-metal bumpers

Parametric Onshape assembly | FIRST Robotics Competition

Designer | Public CAD release: October 2023



TWO-PIECE BUMPER ASSEMBLY | FRAME FROM CAD ANIMATION

150+ teams

using this design

6 inputs

frame, spacing, split and mounting parameters

Quick release

sliding tabs + spring pins

PARAMETRIC DESIGN

I built a configurable two-piece sheet-metal backing assembly in Onshape. Six inputs set frame width and length, bumper spacing, two split openings and mounting height. A second variable studio calculates dependent geometry and updates both parts and assembly mates. Sliding tabs and spring pins provide the mounting interface.

PUBLIC RELEASE AND REACH

I shared the CAD and configuration instructions in the Zebracorns' Chief Delphi build thread in October 2023. More than 150 teams used this design. My contribution was the original configurable design and public documentation.

RELEASE SCOPE

The original model was designed around the 2024 bumper rules. Adaptations for other robots or seasons need mounting-clearance and applicable-rule checks.